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Match activity profile analysis during professional single male pickleball matches

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Abstract

This study aimed to carry out a descriptive analysis of the game actions and activity profile of professional men's singles pickleball. A cross-sectional design was used. Match activity profile of seven male singles pickleball matches from the Carvana Arizona Grand Slam tournament were analysed. Study variables, such as, playing and resting time parameters, shots per rally, type of shots, trajectory of the ball and its path and shot outcomes were studied. The mean match durations were 37.1 ± 11.6 min, with a work-to-rest ratio of 1:3.8 (mean, 0.26 ± 0.03), where the effective playing time represents 20.5% of the total playing time. Most rally durations (77.4%) lasted between 3–9 s and most resting times (76.9%) were between 9–18 s. 53.1% of rallies had between 3–5 shots. The forehand (32.3%), the serve (19.8%) and the backhand (19.5%) accounted for 71.6% of the shots. Cross-court play (55%) predominated over parallel (45%). Most of the shots (99%) were performed from the baseline (53.9%) and the transition zone (45.1%). The majority of the shots were intercepted (80.2%). This information enhances understanding of singles pickleball and can contribute to performance and training session improvements.

Keywords

Intercepted shots, racket sport, rally duration, work-to-rest ratio

Introduction

Pickleball is a sport that comes from the racket-sports family, which combines elements of tennis, padel, badminton and table tennis.[1] Although many people may think that pickleball is a recent creation, it was actually created in 1965, in the United States, by Joel Pritchard; a congressman from Washington State and Bill Bell, a successful businessman.[2] But clearly its popularity has increased within the latest years, making pickleball one of the fastest-growing sports in the USA,[3,4] specifically being massified among older adults from the same country, due to their positive well-being factors.[5] This leads us to the contradiction in which its popularity does not match the number of scientific researches done for getting a better understanding of the sports-specific requirements, players profile and game dynamics. For instance, in the current literature, two investigations can be found regarding technical-tactical performance indicators in single women[6] and in men doubles category[7] that can lead into a jump start for a better understanding of the player's game parameters at a professional level. Although compared to other racket-sports that have been studied and analysed in depth (i.e., tennis), where it has been well established the players activity profile,[8–10] psychological and physiological demands[10–14] in order to guide-training and optimize the tennis players profile. In the same way, padel has articles that deal with the activity profile,[15,16] physiological profile[16,17] and playing actions.[15,18,19] Also, table tennis has been studied at activity profile level and demand profile.[20,21] Comparably, in pickleball, the psychological variables have been studied proving that its practice produces significant improvements in the participants, such as personal well-being, happiness and life satisfaction, among others, and it is concluded that this sport shows potential as a new tool to work on

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and improve people's mental health.[22] Even more, the regular practice can also help to improve cardiorespiratory fitness and positively modify the main risk factors for cardiovascular diseases.[23] Furthermore, studies have been found regarding injuries that can occur in pickleball.[24–26] These conclude that the most typical injuries are knee-related (meniscal pathology, patellar tendinopathy, medial collateral ligament strains and flare-ups of osteoarthritis), shoulders (rotator cuff tendinopathy), achilles tendinitis and plantar fasciitis. In a minor degree, low back pain and hamstring and/or calf strain injuries can be produced. Taking into account the demand profile, only one research was found that concluded that the heart rate for a doubles pickleball game, in a sample with an average age of 65.2 ± 8.0 years, was 108.8 ± 16.7 beats per minute, and the total energy expenditure of a pickleball game was 353.5 ± 85 kcal/game.[23] In view of the players age mentioned before, that there exist a certain limitation within the studied population getting to the result of not knowing accurately the activity profile of competitive players. For this reason and considering the constant growth of this sport, that this study will focus its attention on describing the match activity profile of professional pickleball single male players and their technical-tactical behaviours on court. This information and close-up will deliver valuable indicators for pickleball coaches and a better understanding of the sports-dynamics and game actions to consider in further training sessions.

Methods

Design

An observational cross-sectional design was conducted, being nomothetical and multidimensional, given the plurality of participants and the parameters to be analysed. These ones were classified into various criteria and categories, which are presented in the instrument of observation (Table 1). This was a non-participatory process due to the observer did not interact with the subjects during the observation.[27]

Sample

With the approval of institutional ethics (2324006P), a total of 3668 records were analysed throughout seven males individual pickleball matches corresponding to quarterfinals (n = 4), semifinals (n = 2) and final (n = 1) of the Carvana Arizona Grand Slam tournament held in February 2023. Eight different professional male players were observed (mean ± SD: age, 27.3 ± 5.4 years; height 1.86 ± 0.05 m; background as a professional, 2.6 ± 2.1 years; ranking, 9.6 ± 8.2) recognized by points of the Professional Pickleball Association.[28] Every single player was right-handed for racket manipulation, which does not ensure their dominant upper-laterality.

Table 1. Observational instrument.

Criteria	Categories	Description
Hitting player	SV	Serving player
	RP	Receiving player
Points duration	BP	Beginning of the point
	EP	Ending of the point
Pause between points	BRT	Beginning of resting time
	ERT	Ending of resting time
Player's location	BAS	Baseline
	TRA	Transition
	NVZ	Non-volley zone
Type of shots	FH	Forehand
	BH	Backhand
	FV	Forehand volley
	BV	Backhand volley
	SV	Serve
	SM	Smash
	FL	Forehand lob
	BL	Backhand lob
	FD	Forehand drop shot
	BD	Backhand drop shot
Ball trajectory	CL	Cross-line
	PL	Parallel
Ball actions post hitting	IN	Bounce inside court
	OUT	Bounce outside court
	NB	No bounce
	HNBI	Hits the net and bounces in
	HNBO	Hits the net and bounces out
	HNNB	Hits the net and no bounce
	HNS	Hits the net and stays at the same courtside
	HOP	Hits opponent player
Shot outcomes	INT	Intercepted
	UNF	Unforced error
	FE	Forced error
	WI	Winner

Procedure

Three videos in live format filmed by the PPA and rebroadcast on Youtube through the PPA Tour channel were taken as analysis material. All the observational process was computerized through the LINCE PLUS 1.3.2 software,[29] using an *ad-hoc* observation instrument with specific criteria and categories for the present study (Table 1). The observational instrument was designed following the parameters on previous scientific publications on the analysis of the game and activity profile of tennis, padel and men doubles pickleball[7,9,18] and the official pickleball regulation.[30] It consists of eight criteria, which includes points duration with the beginning and ending time, pause time between points with the beginning and ending of the pauses, player's hitting location on court (Figure 1), type of shot performed, the trajectory followed by the ball, the route of the ball after being hit and the result of the analysed shot. The data from each match was exported in ".csv" format which was

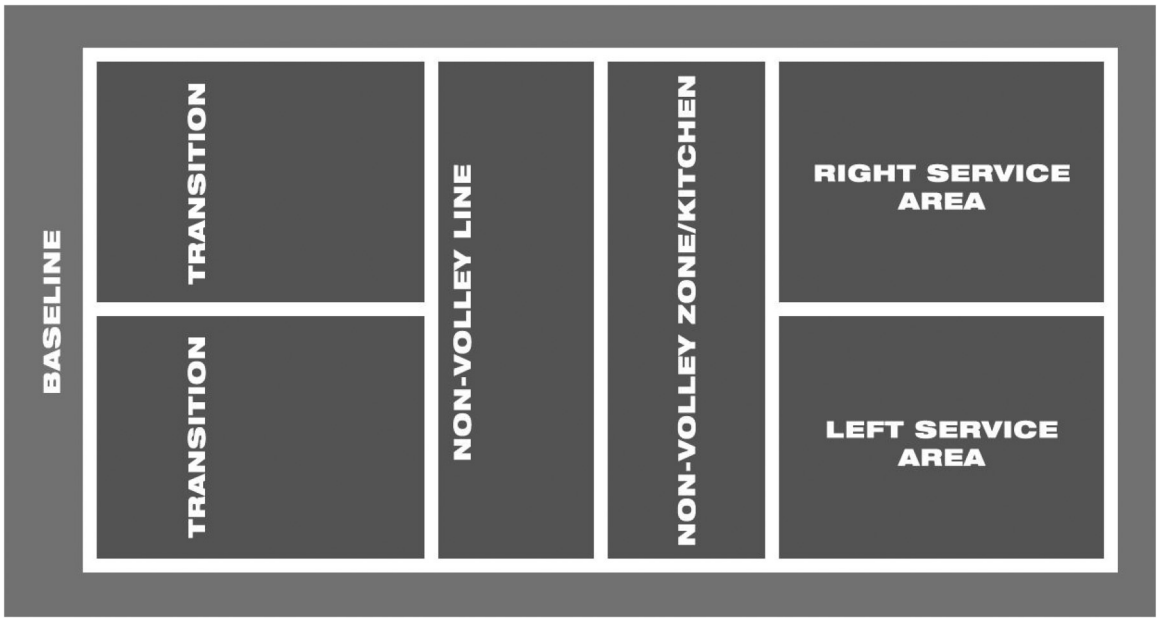

Figure 1. Player's hitting locations on a pickleball court.

transferred to an excel file. From this same excel workbook, all the data processing and analyses were done.

Data quality control

The construct validity was obtained from the consistency of the conceptual framework by four pickleball coaches (graduated from the RFET; Real Federación Española de Tenis – Royal Spanish Tennis Federation), who reached 100% agreement for each of the categories. The validation was carried out using a virtual questionnaire, where the experts could agree or disagree with the categorisation of each criterion. The reliability was determined after the observers had received a period of training in the application of the registration instrument and after reaching an agreement.[31] The reliability of the observation was determined by means of intra and inter-observer agreement in 200 game actions. A 0.91 Kappa coefficient,[32] was obtained in the contrast between the observation carried by the main author and the observation carried out by a racket-sports technician. Same procedure was used for intra-observer agreement resulting in a Kappa coefficient of 0.98.

Statistical analyses

A descriptive study of main study variables was performed. Absolute (n) and relative (%) frequencies were computed for categorical variables, while measures of central tendency and dispersion were calculated for the continuous ones. For the comparison between effective playing time and resting time (in minutes), a paired samples t-test was conducted. The mean difference and 95% confidence intervals were calculated to provide context for the observed differences. For the comparison between rally duration and pause between rallies (in seconds), a non-parametric Wilcoxon signed-rank test was used due to tied pairs and potential non-normality in the data distribution. The Spearman rank-order correlation coefficient for non-parametric data was used to examine the relationship between rally duration and the number of shots per rally. Correlation was classified as trivial (0–0.1), small (0.1–0.3), moderate (0.3–0.5), large (0.5–0.7), very large (0.7–0.9), nearly perfect (0.9), and perfect (1.0).[33] The significance level was set at 0.05. The mean difference and 95% confidence intervals were reported. The reliability was measured with the kappa coefficient.[32] All analyses were performed using version 4.1.3 of the R statistical software.

Results

Time parameters

Figure 2 plots the distribution by intervals of the rally duration and pauses. With a mean really duration of 5.3 s (Table 2), points lasting between 3 and 6 s occurred in 51.9% of the cases, and 25.5% between 6 and 9 s. So that points lasting between 3 and 9 s represents 77.4% of the points played. Points lasting more than 10 s occurred in less than 10% of cases, although there were cases in which the points lasted more than 20 s (0.7%). The longest point recorded was 24 s. On the other end, with a mean resting time between rallies of 20.7 s (Table 2), the pauses duration that occurred most frequently were those lasting between 12 and 15 s (35.2%), between 9 and 12 s (27.2%) and between 15 and 18 s (14.5%). So that resting times lasting between 9 and 18 s represents 76.9% of the

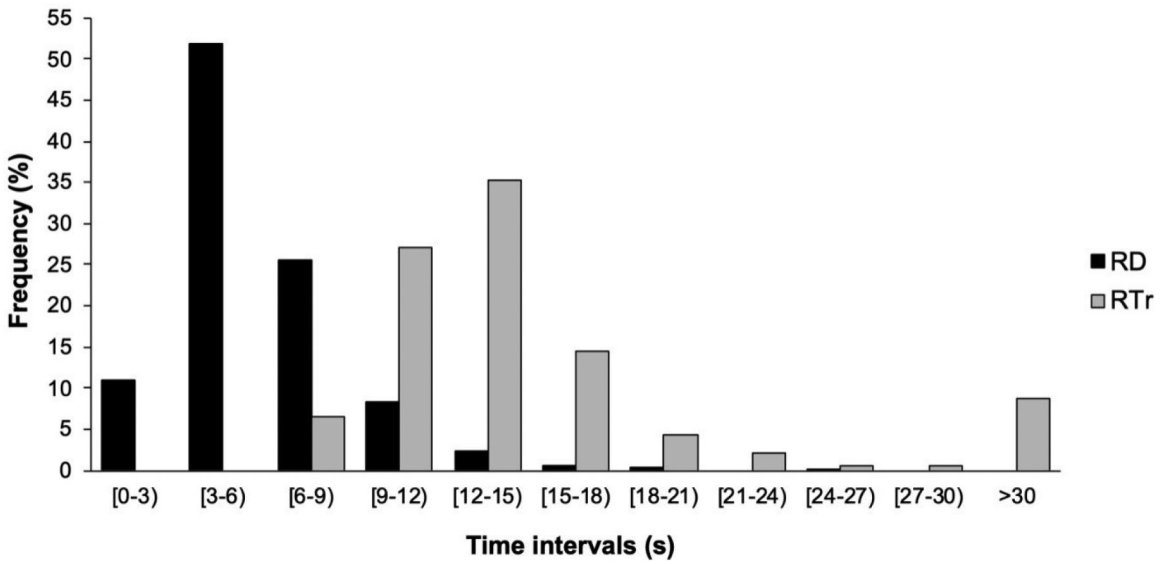

Figure 2. Playing time (rally duration) and recovery (pauses between rallies) intervals.

Table 2. Time parameters of a singles male pickleball match.

	Mean (SD)
RD (s)	5.3 (3.0)
RT (s)	20.7 (20.1)
EPT (min)	7.6 (2.1)
Pause (min)	29.5 (9.6)
Total match time (min)	37.1 (11.6)
Work-rest ratio	1:3.8 (0.5)

RD, rally duration; RT, resting time between rallies; EPT, effective playing time.

case. However, there were a considerable number of pauses lasting more than 30 s (8.8%). The shortest pause recorded was 6 s and the longest was 253 s. The average time of an individual men's pickleball match was 37.1 ± 11.6 min with and effective playing time of 20.5% of the total time of a match and a work-to-rest ratio was 0.26 ± 0.03, in other words, 1:3.8 (Table 2). The comparison between effective playing time and resting time, as well as rally duration and pause between rallies, showed significant differences ($p < .001$). The mean differences and 95% confidence intervals were 21.8 min (14.8 to 28.9 min) for effective playing versus resting time, and 15.1 s (12.8 to 17.5 s) for rally duration versus pauses between rallies.

Points and shots

A total of 3060 shots (mean per game, 437.1 ± 124.7 shots) and 607 points (mean per match, 86.7 ± 29.2) have been analysed, in which only 283 (mean per match, 40.4 ± 13) have finally hit the scoreboard. Figure 3 shows the distribution of shots per rally in which the 53.1% of the rallies consisted in 3–5 shots. The shortest rallies noted only 1 shot (4.4%) and there were few cases in which the interchange consisted in more than 13 shots (2.1%). A significant, very large positive correlation was found between between rally duration and the number of shots per rally ($r = 0.891$, $p < .001$, 95% CI = 0.873–0.907).

Type of shots and ball trajectories

The forehand (32.3%; mean per match, 141.3 ± 60.5), the serve (19.8%; mean per match, 86.5 ± 29.8) and the backhand (19.5%; mean per match, 85.1 ± 25.9) account for 71.6% of the shots (Table 3). The backhand (14.2%; mean per match, 62 ± 23.6) and the forehand (8.5%; mean per match, 37.3 ± 11) volleys represents the 22.7 of the shots. According to the ball trajectories, the cross-line (55.1%; mean per match, 240.8 ± 60.8) had a predominance over the parallel (44.9%; mean per match, 196.3 ± 72.3).

Players hitting locations

Almost all of the of the shots performed (99%) were made from the baseline area (53.9%; mean shots per match, 235.9 ± 75.1) and the transition zone (45.1%; mean shots per match, 197 ± 60.7). In the non-volley zone, also known as the kitchen, very few shots occurred (0.1%; mean shots per match, 4.3 ± 1.9) (Table 3).

Shot outcomes

The majority of shots ended up being intercepted (80.2%; mean shots per match, 350.5 ± 98.4). Far behind are unforced errors (8.7%; mean shots per match, 37.9 ± 17.5), winning shots (6.2%; mean shots per match, 27 ± 6.1) and forced errors (5%; mean shots per match, 21.7 ± 8.2) (Table 3).

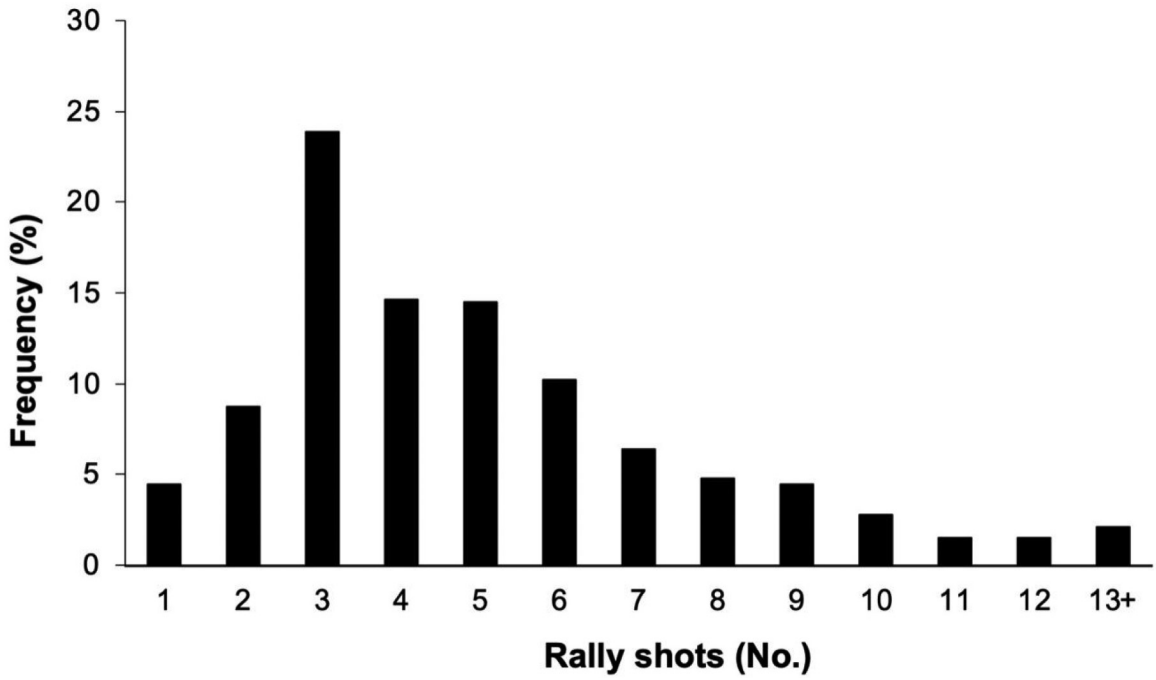

Figure 3. Distribution of shots per rally.

Post-hitting ball actions

Mostly, the ball bounced inside the opponent's field (60.1%). In the case in which the ball did not bounce (i.e., it was hit directly by the opponent), are less frequent (24.8%). Besides these values, shots were found when the ball ended up bouncing outside (7.2%) and when it hit the net completely, staying in its own field (6.3%) (Table 3). Other shots were not usually given.

Discussion

To our knowledge, this is the first study that provides detailed information on playing actions and activity profile of professional males individual pickleball based on the analysis of the characteristics of each shot and duration parameters of the points. The results showed that the mean duration of the point was 5.3 ± 3 s interposed with pauses of 20.7 ± 20.1 s with a total effective playing time of 20.5%. The points mostly lasted between 3 and 9 s (77.4%) whereas pauses between 9 and 18 s (76.9%). 53.1% of the rallies consisted in 3–5 shots. The forehand (32.3%), the serve (19.8%) and the backhand (19.5%) accounted for 71.6% of the shots, with a predominance on cross-line ball trajectories (55.1%). The player's common hitting locations were the baseline area (54%), followed by the transition zone (45.1%). According to the International Pickleball Federation, there is a rule that only points are added to the scoreboard when points are won while serving, therefore, the number of points per game can be very variable, and consequently, the number of shots and the duration of the games as well. Hence, a total of 607 (mean per match, 86.7 ± 29.2) points have been analysed, in which only 283 (mean per match, 40.4 $\pm$ 13) have finally hit the scoreboard.

The average points duration was 5.3 ± 3 s, lasting the majority of the points between 3 and 6 s (51.9%), showing a difference with the average of points duration on men doubles category (10.7 ± 8.7 s) where the 55.4% of the points lasted between 3 and 9 s.[7] Comparing these results with tennis, we could see that there are higher percentage of points that end in less than 3 s, although the average duration of the points is slightly longer than pickleball points.[8,10,34,35] Taking into consideration, in pickleball it is very difficult to produce winning serves, while in tennis it is much easier (aces). However, points that end as quickly as a direct serve in tennis can also come out, because there is no second serve, therefore, if the first serve is missed, the point is over, although this situation is not very frequent at a professional level.[6,7] Likewise, padel has a longer duration including values from 7.24 ± 0.0 to 16.8 ± 2.8 s,[16] considering the frequent use of lobs and wall-bounces during the game, which could be a reason why longer rally durations are produced.[36] Comparing the duration of the points with table tennis, we see how in this sport the points tend to last less than in pickleball.[20,37] The interval of frequency of resting time duration is quite similar to the given in tennis, although the average pause time between points is somewhat lower.[8] In pickleball there are pauses lasting more than 30 s (8.8%), which usually represents the

Table 3. Technical – tactical parameters during a single male pickleball match.

	n	%
Types of shots		
Forehand	989	32.3
Serve	606	19.8
Backhand	596	19.5
Backhand volley	434	14.2
Forehand volley	261	8.5
Backhand drop shot	67	2.2
Smash	35	1.1
Backhand lob	25	0.8
Forehand drop shot	25	0.8
Forehand lob	22	0.7
Players hitting locations		
Baseline	1651	53.9
Transition	1379	45.1
Non-volley zone	30	1
Shot outcomes		
Intercepted	2454	80.2
Unforced error	265	8.7
Winner	189	6.2
Forced error	152	5
Ball actions post-hitting		
Bounce inside court	1838	60.1
No bounce	759	24.8
Bounce outside court	221	7.2
Hits the net and stays at the same courtside	193	6.3
Hits the net and bounces in	33	1.1
Hits the net and no bounce	8	0.3
Hits the net and bounces out	4	0.1
Hits opponent player	4	0.1

N, absolute value; %, relative value.

pauses given when calling timeout (1 min) or at the end of a set (2 min). The most repeated rally was the 3-shot interchange (23.9%) followed by 4 (14.7%) which resembles with the value found in tennis with a mean of 4.1[10] albeit being far below compared to padel were the mean is 9.5 shots per point.[36] Regarding rally durations and shots per point, a very large positive correlation was found between these two variables which was completely expected, as rally duration gets longer the more shots per points were performed. Compared to pickleball men doubles category, results showed that in 20.1% of the cases, rally exchanges consisted in 5–6 shots, existing a difference due to the number of players sharing the court (four instead of two).[7] Match durations represented an average of 37.1 ± 11.6 min with a work-to-rest ratio 1:3.8, which is similar to the timing on table tennis and badminton matches,[37–39] having differences than the duration of a tennis match.[10,40,41] By other end, there exists a difference compared to padel, where the work-to-rest ratio notes a higher value due to the extended rally durations.[16] Given the significant differences noted between effective playing and resting time, as well as rally duration and pause between rallies ($p < .001$), it is an important parameter to consider when designing further training programmes.

Just like in pickleball men doubles (40.2%)[7] and tennis,[9,42,43] the forehand was the most performed shot (32%). The predominance over the backhand (19.5%), could be because in many points, when players noticed a centred-ball, they tended to reverse themselves for better shot performance (skilled face of the racket). These results differ with other racket-sports (i.e., padel) where the most performed shots are the volleys.[18,44,45] The backhand volley (14%) showed a higher percentage than the forehand volley (9%) because, during a point, there are quick interchanges from a very close distance between players, which means a very short time interval to react, and in these cases, the backhand volley acts as a "shield" covering more body than the forehand volley for defending. These results also are repeated at men doubles category, where the backhand volley outnumbered the forehand volleys (18.4% and 12.7% respectively).[7] There were not many lobs (1.5%), possibly due to the court length is not very long, therefore, whenever performed, it was a last resource tool. And, as for the drop shots (3%), likewise as the lobs, as the court is not long, the fact of performing one of these shots is not usually considered as an option to make a winning shot, rather is usually used to approach the opponent player to the net and then generate a winning shot or a forced error.

Regarding the player's hitting location on court, there was a predominance of the baseline shots (54%) over the transition (45.1%) which is the zone that follows the baseline towards the non-volley zone. Same situation occurred at women singles in which the baseline zone accounted more shots performed rather than the transition (38.6% and 35.8% respectively).[6] Interesting situation happened at men doubles, where opposite results were found in which shots performed at the transition zone (60.7%) exceeded those performed at the baseline (15.2%).[7] This notes that players at individual categories, intend to balance their game play prioritizing the baseline and the transition zone over the non-volley zone; where the fewer shots are performed due to the limitations of the rule of the non-volley zone, which does not allow to enter this zone to hit the ball, unless it has bounced before. Moving forward, during an attacking phase, it seems that players are more interested in hitting the ball in a direct way from the transition limit zone with the non-volley line, rather than the ball bouncing inside the non-volley zone and then hitting the ball in an indirect way. Performing a shot in an indirect way supposes a greater interval of time between interchange of the ball, which gives more responding time to the opponent. On the contrary, when a player is being dominated by the opponent, it is in his interest to perform a shot so that it bounces inside the contrary non-volley zone, in order to have more time to recover the position.

As for the shot outcomes, 80.2% of the shots were intercepted, 8.7% unforced errors, 6.2% winners and 5% forced errors. These percentages obtained are quite different from those of tennis, where the winning shots and unforced errors are more common.[35,42,46] In contrast, the percentages obtained in the present study agree with those found in men doubles category[7] and padel.[18] In relation to the ball trajectory, the fact that the cross-line (55%) game was predominant over the parallel (45%), is definitely related to the sport-regulations, which says that serves must be performed in a cross-line. In padel, we could find a study that makes the same reflection regarding this aspect, in which cross-line balls predominate over the parallels.[18,47] Regarding the ball actions after being hit, the bounce predominated (60.1%) over the non-bounced balls (24.8%). This gives us a bigger number of indirect shots over the direct ones. Taking into account the pickleball regulation that describes the bounce rule, in which the serve must bounce in the opposite court before hitting, that these differences between indirect and direct shots were totally expected.

It should be noted that the obtained results of this research might be useful when establishing parameters and criteria on future training sessions for players competing at single category. Regarding technical-tactical aspects, it is primordial to consider the fact of practicing every shot during training sessions, emphasizing the most performed ones: forehand, backhand and serve, especially this last one, which requires full attention as it is a "single-shot" shot. Hitting locations are also recommended to work on, specifically from the baseline as well as from the transition zone and favouring the cross-court patterns. In addition, the percentage of effective playing time, points durations and the number of shots per rally, allows coaches and players to approximate the workload of matches into practical sessions, which is essential for developing the player's performance.

This study represents an initial attempt to understand the patterns and behaviour of professional male players in match situations. Although the sample consists of only eight players and seven matches, the analysis includes a total of 3668 recorded shots, providing a detailed insight into performance trends. Nevertheless, the findings should not be generalised beyond this specific context. Future studies should expand the sample to include players of different skill levels, ages, and genders, as results may vary accordingly. Additionally, this study focused exclusively on technical-tactical performance indicators, without considering physiological or physical demands, which would be an interesting avenue for future research.

Conclusion

Understanding the competitive dynamics of pickleball is of vital importance in its daily practice, and coaches should know the characteristics of the match actions that occur and the activity profile of the sport in which they are involved. The mean duration of a competitive match is about 37.1 min, with a work-to-rest ratio of 1:3.8, where the actual playing time represents almost 21% of the total playing time. Most of the points last between 3 and 9 s and the pauses between 9 and 18 s. Shots per rally are commonly consisting in 3 to 5 shots. The most frequent shots in professional males individual pickleball are the forehand, serve and the backhand. There is a predominance of the cross-line game over the parallel. Almost all of the shots are performed from the baseline and the transition zone. Intercepted shots are the most produced during a match rather than unforced errors, winning shots and forced errors. The majority of the ball post hitting actions are the bounces over the non-bouncing balls which determine more indirect shots rather than direct ones.

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